

SERGEY ZINCHENKO



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Open to remote / relocation worldwide



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R&D engineer with 3+ years building learned systems — Huawei's query optimizer (+50% over SotA) and recent work on LLM training and efficient deep learning. Strong mathematical foundations (Olympiad medalist, MSc with honors). Publications at ICML Workshop and in Neural Networks (Q1).

EXPERIENCE

PRESENT 2025	Independent Engineering DL/LLM Engineer	Remote
Self-directed work on LLM training, fine-tuning, and efficient inference. Built a [cost model] predicting peak memory and compute from LLM architecture and kernel choices to within 1% of measured (validated on DDP/FSDP with Flash Attention, Triton, torch.compile, Liger). Implemented [model parallelisms] (TP, SP, PP) for LLaMA on torch.distributed. Trained Qwen2.5-1.5B on GSM8K math reasoning with [GRPO + LoRA] in pure PyTorch.		
2025 2022	Huawei R&D Engineer	Novosibirsk, Russia (on-site)
Led development of [HERO]: a learned query optimizer achieving 50% query latency reduction over prior SOTA, with 6–17x faster inference resolving the deployment bottleneck. Designed and implemented the core components to meet the interpretability and reliability bar that kept earlier learned optimizers <i>out of production</i> . Invented a hybrid training approach: an <i>error-prone</i> NN accelerates training by guiding search, while the deployed model is lightweight, inspectable, and reliable. Designed and implemented a [benchmark framework] that cut experimentation time from days to minutes by replaying cached query executions instead of re-running them. Proposed a [specialized data structure] over sets of trees that accelerated a core optimization stage in SOTA analytical workload optimizers, including Microsoft's framework and RL-based pipelines.		
2022 2021	HSE (QS Top 50 in Math; HDI lab) DL Researcher	Novosibirsk, Russia (remote)
Invented and implemented [SnapStar] - a lightweight ensembling method matching the accuracy of large neural networks while training only small ones. Using statistical learning theory, proved its <i>optimality</i> and <i>robustness</i> , with results published in a Q1 journal and an ICML workshop.		

PUBLICATIONS

arXiv, 2024	HERO: Hint-based efficient and reliable query optimizer [paper]
arXiv, 2024	The selection problem in MQO: a comprehensive survey [paper]
NN Journal (Q1), 2024	Star algorithm for neural network ensembling [paper]
ICML Workshop, 2022	SnapStar algorithm: a new way to ensemble neural networks [paper]

EDUCATION

2024 2022	Master's Degree in Applied Math (with honors; 4.957/5.0) Master's Thesis: "Hint-based query optimization for databases"	NSU (QS Top 110 in Math, 2024) Russia
COURSES: Database Optimization, Deep Learning		
2022 2018	Bachelor's Degree in Math and Computer Science (with honors; 4.975/5.0) Bachelor's Thesis: "Neural network ensemble algorithm"	NSU (QS Top 117 in Math, 2022) Russia
COURSES: Statistical Learning, Advanced Algorithms and Data Structures		

ACTIVITIES / ACHIEVEMENTS

Olympiads	Gold Medalist, International Olympiad in Cryptography (NSUCRYPTO), 2023 1st Degree (top 27/1,700+), National Student Math Olympiad ("I Am Professional"), 2022
Contests	Ranked among top 1% worldwide in LeetCode contests
Awards	"Employee of the Year" (Huawei, 2023), "Hope of the Faculty" (NSU, 2019)
Talks	Lecture on HERO at ISPRAS Open, 2024 and [master class] at PGBootcamp, 2023

TECHNICAL EXPERTISE

Stack	Python, PyTorch (multi-gpu), HuggingFace, Triton (fused kernels), NumPy, Pandas, scikit-learn, SQL
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